

Applying Computer Vision to Analyze Self-Injurious Behaviors in Children with Autism Spectrum Disorder

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Abstract

Computer vision has immense potential to advance healthcare and behavioral research, particularly for conditions such as autism spectrum disorder (ASD), which is characterized by early-emerging social communication deficits and repetitive sensory-motor behaviors. Early detection of these behavioral patterns is crucial for timely diagnosis and intervention. While existing studies primarily focus on facial expressions and eye-tracking in children with ASD, the study of body gestures remains underexplored, largely due to the lack of datasets capturing full-body movements. In this work, we present a novel dataset of videos capturing full-body gestures in children with ASD, recorded during behavioral therapy sessions. The dataset specifically includes frames capturing severe behaviors such as head hitting and head banging. To establish benchmarks, we conducted baseline experiments using state-of-the-art image classification models on these action-specific frames. Furthermore, we evaluated kinematic pose estimation models on the same frames, analyzing their performance and highlighting the unique challenges faced in applying computer vision techniques to children with ASD. This dataset and our findings aim to bridge critical gaps in current research, providing a foundational resource and insights for advancing ASD-focused behavioral assessments through computer vision.

1. Introduction

In recent years, computer vision has emerged as a transformative tool in healthcare, enabling non-invasive, accurate, and timely assessments for various conditions, including neurodevelopmental and behavioral disorders. For children diagnosed with autism spectrum disorder (ASD), the potential of computer vision in behavioral studies is particularly promising. ASD is a neurodevelopmental disorder characterized by deficits in social communication and the presence of repetitive behaviors,



Figure 1. Representative frames showcasing three distinct self-injurious behaviors (SIB) labeled during multi-modal analysis across different functional analysis (FA) sessions. Each frame is annotated with predefined behavioral definitions, illustrating key events including head banging, hitting the head with left hand, hitting the head with right hand.

such as self-injurious behaviors (SIB) like head banging or self-hitting [3]. By processing and analyzing video data at scale, computer vision offers a unique opportunity to identify and monitor these behavior patterns with objective and quantifiable metrics.

Much of the existing research in autism diagnostics has focused on analyzing facial expressions, eye gaze, and other visually observable cues. For example, eye-tracking studies have shown that gaze patterns can signal neurodivergence, providing a basis for early diagnosis of

ASD. While these visual indicators are invaluable, they represent only a subset of the complex behavioral repertoire of children with ASD. Equally critical for diagnosis and intervention are body gestures, such as repetitive movements and self-stimulatory behaviors, which remain underexplored in the domain of autism-focused computer vision research.

The limited focus on body gestures can be attributed to a lack of datasets capturing full-body movements in children with ASD. Most available datasets emphasize close-up facial expressions or eye-tracking data, with few providing the breadth needed to analyze diagnostically relevant body gestures. Addressing this gap, we present SIB-ASD, a novel dataset specifically designed for studying intense self-injurious behaviors in children with ASD. The dataset includes over 8,500 annotated frames capturing behaviors such as head hitting (with either hand) and head banging. Data were collected during functional analysis (FA) sessions in a controlled clinical environment, ensuring consistency and reliability.

Our work makes the following contributions:

- **A novel dataset:** The SIB-ASD dataset is introduced, providing a focused resource for analyzing self-injurious behaviors in children with ASD. The dataset includes annotated frames highlighting diagnostically significant actions, with example snapshots presented in Figure 1.
- **Action Detection:** We conduct baseline experiments on SIB-ASD using state-of-the-art (SOTA) image classification models. These benchmarks establish a foundation for further exploration of algorithms tailored to ASD action classification.
- **Pose Estimation:** We evaluate SOTA kinematic pose estimation models on the designated frames and provide a detailed discussion on the challenges encountered when applying computer vision techniques to study behaviors of children with ASD.

By addressing the scarcity of datasets and establishing robust benchmarks, this work aims to advance the application of computer vision in monitoring critical behaviors in children with ASD, ultimately contributing to more effective diagnostic and therapeutic strategies.

1.1. Small Data Statement

This research qualifies as small data research due to the ethical and logistical constraints of working with sensitive populations, such as children diagnosed with ASD, and the challenges of capturing rare but diagnostically significant behaviors like SIB. The dataset presented in this study, SIB-ASD, is specifically designed

for analyzing SIB in children with ASD. Data were collected during FA sessions conducted in a controlled clinical environment, ensuring consistency and reliability. The dataset comprises of 8,557 annotated frames with three distinct behaviors: head hitting with the left hand, head hitting with the right hand, and head banging.

To ensure robust ground truth labeling, annotations were validated using synchronized accelerometer data, providing a high degree of accuracy. Videos were recorded with participants' consent under strict privacy-preserving protocols, balancing ethical considerations with the need for comprehensive data. Despite these measures, the dataset remains inherently limited in size due to the challenges of collecting and annotating behavior-specific data within this context.

To address these limitations, we adopted several strategies to enhance the dataset's utility and impact. The controlled clinical setup during FA sessions ensured consistency in data collection, reducing variability and improving reliability. Additionally, we employed transfer learning by utilizing state-of-the-art pre-trained models for image classification and kinematic pose estimation, leveraging knowledge from larger datasets to improve performance on the limited data. Baseline experiments were conducted using SIB-ASD to evaluate the performance of image classification and pose estimation models, establishing a benchmark for future research and addressing key challenges in applying computer vision techniques to ASD-focused studies.

By combining a controlled setup, robust validation methods, and advanced transfer learning techniques, this work demonstrates how small, carefully curated datasets like SIB-ASD can effectively address gaps in ASD-focused research, enabling meaningful advancements in the study of self-injurious behaviors.

2. Related Work

Computer vision techniques have been increasingly applied to understand ASD, focusing on symptom detection, pose estimation, and activity recognition [12, 18]. However, existing approaches face challenges in addressing the unique complexities of ASD behaviors and in utilizing robust datasets [3, 6]. Recent studies have explored the challenges of monitoring self-injurious behaviors in individuals with ASD through technology, highlighting the importance of addressing unique user needs and the limitations of current approaches [3]. Furthermore, video-based recognition of infant actions, which is a critical aspect of early autism detection, presents significant challenges in terms of dataset availability, annotation quality, and algorithmic reliability, as critically examined in [6]. With that said, existing literature on computer vision oriented studies in the field of

ASD can be classified into three categories: studies that create ASD-related datasets and baselines, studies Applying Pose Estimation on ASD data, studies applying action recognition on ASD data.

2.1. ASD-related Datasets and Benchmarks

Notable datasets have been introduced in recent years to advance computer vision research in ASD. For instance, CALMED is a multimodal annotated dataset tailored for emotion detection in children with autism, incorporating diverse behavioral cues to facilitate deeper understanding of ASD-related expressions [22]. Similarly, the “Hear Me, See Me, Understand Me” dataset [4] combines audio and visual modalities to enable behavior recognition in individuals with autism, addressing the need for integrated approaches to capture multimodal signals. [20] introduces another novel dataset designed for video-based autism classification that leverages extra-stimulatory behavior, providing a unique resource for analyzing subtle ASD characteristics. Additionally, the Stimming Behavior Dataset unifies stereotypical behavior data collected in naturalistic environments, offering a valuable resource for action recognition and behavior analysis in the context of ASD [17].

2.2. Pose Estimation on ASD data

Pose estimation techniques have been applied by a number of studies to detect and understand atypical behaviors in individuals with ASD. For instance, [14] explored the detection of arm flapping in children with autism using human pose estimation and skeletal representation algorithms. Similarly, [21] utilized facial expression recognition and head pose estimation to facilitate the early diagnosis of ASD, showcasing the improved diagnostic accuracy by integrating emotion recognition. [16] developed a gesture recognition system for autistic children using person pose estimation and supervised learning, providing a framework for analyzing and interpreting gestures in therapy sessions. Additionally, [27] presented a method to recognize atypical behavior for autism diagnosis by applying pose estimation over time, emphasizing the importance of temporal dynamics in capturing nuanced behaviors.

2.3. Action Recognition on ASD data

Action recognition techniques assist in analyzing ASD-related behaviors, for monitoring and detecting stereotypical motor movements. Early efforts, such as [1], utilized decision trees and rule-based algorithms to classify the observed behaviors within laboratory and classroom settings to recognize such movements. Similarly, [5] developed an activity recognition framework based on joint estimation, demonstrating the effectiveness of skeletal data for ASD-related movements.

More recent studies such as [28] tackled the challenge of detecting autism-related headbanging using time-distributed convolutional neural network with limited training data from moving cameras. Another innovative approach by [11] combined a parent-child dyads block-play protocol with a hybrid deep learning framework to improve early detection of ASD, emphasizing the role of structured interactions in behavior analysis. Vision-based recognition systems have also made significant contributions to this domain. [29] proposed a framework for recognizing autism-related behaviors through computer vision, highlighting its potential for unobtrusive monitoring. [15] introduced a vision-assisted system that employs convolutional neural networks and a bag-of-visual-words model for early diagnosis of ASD through stereotypical behavior recognition, addressing the scalability and precision for clinical applications. With all the endeavors made in the mentioned fields, computer vision has yet to reach full potential in research on ASD. This gap is primarily due to challenges in data gathering and publishing processes, including privacy concerns and difficulties with controlled data-capturing settings. In this work, we introduce a lean dataset captured in a controlled environment with planned, relevant activities. By obtaining proper consents and employing privacy-preserving methods, we aim to address the critical issue of data availability. Furthermore, we benchmark state-of-the-art (SOTA) image classification and pose estimation models on our dataset to provide a foundational starting point for leveraging its beneficial features.

3. Methods

By creating the dataset and the experiments we aim to contribute to the studies on children with ASD by overcoming the challenges stemming from the unavailability of data. We are also looking to achieve an action recognition and kinematic pose estimation benchmark for a few of the most prevalent repetitive intense behaviors in children with Autism. This section begins by detailing the process of data capturing, data preprocessing, and then delves into the implementation of image classification and pose estimation models.

3.1. Data Capture

We present a dataset called SIB-ASD as a benchmark collected from videos captured from 2 children, one Hispanic male of age 14 and one Hispanic female of age 9. The data was recorded during FA sessions conducted in a controlled environment within a university-run outpatient healthcare clinic. The sessions were held in treatment rooms equipped with worktables, lounging furniture, and age-appropriate stimuli tailored to each partic-

ipant’s preferences. This research was approved by the Institutional Review Board (IRB), and caregivers provided informed written consent prior to participation. The overall study, however, was conducted with a total of 12 children, but only 3 finished the study with 2 consenting to have their data used for such research purposes. The SIB-ASD dataset includes multi-modal data of SIB events, offering synchronized accelerometer readings and labeled video frames. Frame-by-frame annotations were conducted using predefined behavioral definitions, capturing the start and end times of each SIB event with precision. 2 raters reviewed the data separately with disagreements calibrated among them to generate a perfect set of annotations for each video frame. These annotations, validated against accelerometer data, establish a robust ground truth dataset for testing and evaluating behavioral event recognition models. While the dataset includes frame-level labels for specific behaviors, it does not contain keypoint annotations required for quantitative pose estimation tasks. This limitation emphasizes the need for future datasets to incorporate keypoint-level annotations to facilitate supervised pose estimation approaches and expand research applications. Additionally, the dataset encompasses diverse behavioral topographies and experimental conditions across participants, enabling its application in developing advanced automated analysis tools for SIB detection. Statistical analyses, including inter-observer agreement, sensitivity, and specificity measures, underscore the reliability and richness of the dataset. These features make SIB-ASD an invaluable resource for advancing both behavioral event recognition and the automated analysis of SIB in clinical and research settings.

Data Preprocessing: Frame extraction, frame labeling, and resolution standardization are the key components of our preprocessing pipeline, enabling efficient preparation of data for SIB activity detection task. The pipeline processes input video sequences captured during functional analysis (FA) sessions, generating labeled frames and preparing them as input to state-of-the-art models. Input video V is represented as a sequence of T frames, $V = f_1, f_2, \dots, f_t, \dots, f_T$ captured from two subjects diagnosed with ASD. From the extracted frames, a subset corresponding to specific SIB events was identified, amounting to 8,557 frames labeled as hit head with right hand, hit head with left hand, and head bang. Frames not corresponding to these labels were discarded to create a focused dataset of relevant behavioral events. We used a 80-20 train-test split, with 20% of the frames of each label randomly selected for testing to ensure balanced representation between classes. This split aligns with standard machine learning practices and ensures consistency across experiments, facilitating fair benchmarking and reliable evaluation of model performance.

To ensure compatibility with various deep learning models, the labeled frames were resized using LANCZOS resampling method, chosen for its effectiveness in preserving visual quality during resolution reduction. The resulting frame resolutions were standardized.

3.2. Activity Detection

3.2.1 Problem Statement

The goal of this study is to develop a robust training and evaluation pipeline for detecting one of three distinct activities — head hitting with the left hand, head hitting with the right hand, and head banging—from a given input image. Of the 8,557 total images, 1,623 are labeled as head banging, 2,553 as hitting head with the left hand, and 4,496 as hitting head with the right hand. From each category, 20% of the images are selected randomly as the test set. The task of activity detection is performed through a simple classification approach wherein we fine-tune pre-trained deep learning models, adapting their architectures to the dataset, and optimize performance through effective training strategies, including auxiliary outputs and efficient GPU resource management.

3.2.2 Methods Evaluated

The training and evaluation pipeline is designed to support various state-of-the-art deep learning models. Computations are performed on a GeForce RTX 4080 GPU, with input data structured into a batch size of 16. Images are resized to 224×224 pixels for all models except Inception_V3, which requires a resolution of 299×299. Training and validation datasets are organized into separate directories for streamlined access during iterations. The pipeline incorporates model architecture adjustments tailored to the dataset’s three classes. Specific modifications were implemented to ensure compatibility of the pre-trained models with our data. All models leverage pre-trained weights from ImageNet, utilizing the rich feature representations learned from large-scale image data. This transfer learning approach allows for efficient adaptation to ASD-specific tasks despite the relatively small dataset size. For models such as Inception_V3 and GoogLeNet, the fully connected (fc) layer was replaced to match the number of classes, and auxiliary outputs were incorporated by modifying auxiliary layers where available. In architectures like DenseNet121 and EfficientNet_V2_S, the classifier layer was updated to include a new final fully connected layer, while MobileNet_V3.Small required modifying the third element in its classifier. For ResNet101 and Wide_ResNet101_2, the fc layer was replaced, while Vision Transformer (ViT) model required adjustment

Method	Parameters	FLOPs	Accuracy	F1 Score	Precision	Recall	AUC
Mnasnet [25]	3.1	0.32	0.9781	0.9778	0.9774	0.9781	0.9962
Regnet [19]	22	8.4	0.9783	0.9792	0.9801	0.9783	0.9971
DenseNet121 [9]	8	2.88	0.9820	0.9808	0.9797	0.9820	0.9982
EfficientNet_V2_S [26]	22	8.4	0.9783	0.9781	0.9778	0.9783	0.9964
GoogleNet [23]	7	1.43	0.9779	0.9782	0.9786	0.9779	0.9970
Inception_V3 [24]	23.8	5.72	0.9829	0.9817	0.9805	0.9829	0.9973
MobileNet_V3_small [8]	2.5	0.22	0.9777	0.9772	0.9768	0.9777	0.9942
ResNet101 [7]	44.5	7.6	0.98	0.9811	0.9822	0.9778	0.9977
ViT_b.16 [2]	86	17.6	0.8881	0.8905	0.8947	0.8881	0.9704
Wide_ResNet101.2 [30]	126	23.1	0.9772	0.9772	0.9772	0.9772	0.9967

Table 1. Results, parameters ($\times 10^6$) and FLOPs ($\times 10^9$) of 10 SOTA image classification models on our dataset. We use the pretrained weights for each model from the default implementation in Pytorch library.

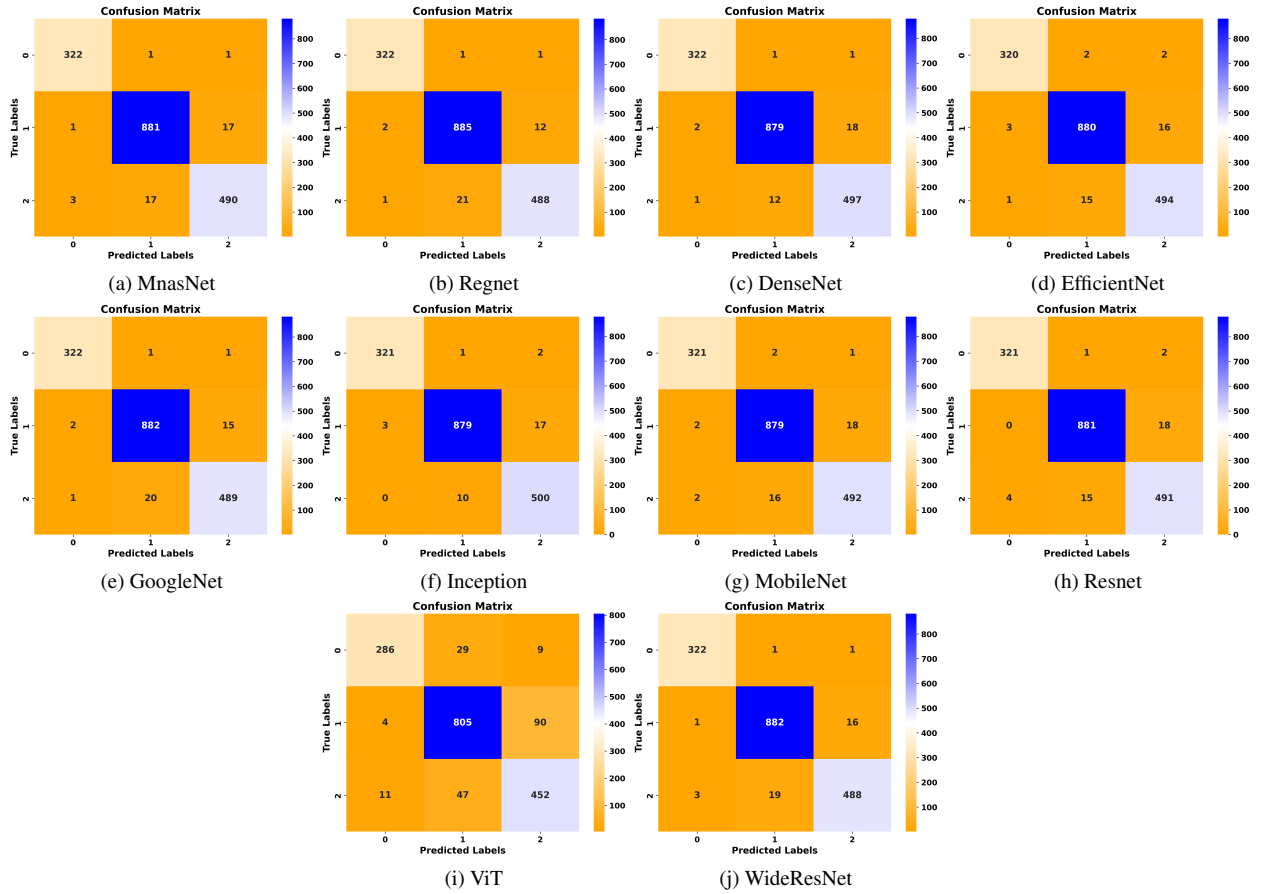


Figure 2. Confusion matrices of models fine-tuned on our dataset with the encoded three labels with 0 representing hitting head with the right hand, 1 representing hitting head with the left hand, and 2 representing banging head on other objects.

of their head or heads.head attributes for classification. During training, image-label pairs are processed in batches, predictions are computed using forward passes, and weights are updated via backpropagation. The primary loss function is CrossEntropyLoss, with additional weight assigned to auxiliary losses when applicable.

Utility functions save model weights after each epoch, retaining the best-performing weights based on validation loss. GPU memory management and resource utilization are tracked using NVIDIA's NVML, ensuring optimal hardware usage across models. This efficient pipeline design allows for flexible adaptation to diverse

architectures and provides a robust comparative framework for evaluating state-of-the-art models.

3.2.3 Results

As Table 1 and Figure 2 indicate, the performance of image classification models was evaluated using metrics such as accuracy, F1-score, precision, recall, and AUC, with confusion matrices providing detailed insights. DenseNet121 achieved the highest accuracy (98.20%) with an F1-score of 0.98 and an AUC of 0.998, correctly classifying 1,698 out of 1,733 images, including 322/324 for “head hitting with the right hand,” 879/899 for “head hitting with the left hand,” and 497/510 for “head banging.” Similarly, Inception_V3 showed excellent performance with an accuracy of 98.29% and an F1-score of 0.98, accurately predicting 1,700 images, including 321/324, 879/899, and 500/510 for the respective classes. EfficientNet_V2_S (97.83% accuracy) and GoogleNet (97.79% accuracy) performed comparably, though they exhibited minor weaknesses in distinguishing “head banging” behaviors, correctly classifying 1,694 and 1,693 images, respectively. Other CNN-based models, like ResNet101 (98.00% accuracy) and MobileNet_V3_Small (97.77% accuracy), demonstrated robust overall performance, but struggled slightly with overlaps between “head hitting” classes, confusing the right with the left hand. In contrast, ViT_B_16 showed the weakest performance, with an accuracy of 88.81% and 1,543 correctly classified images, struggling particularly to separate “head hitting” from “head banging.” These results underline the strength of CNN-based architectures for ASD-related activity classification, with models like DenseNet121 and Inception_V3 leveraging their spatial hierarchies for superior performance. The confusion matrices highlight challenges in reducing misclassifications in the “head banging” class, suggesting that future work could integrate temporal or multi-modal data to enhance classification reliability. Overall, the CNN models provided a strong benchmark, demonstrating their capability to generalize across the dataset.

3.3. Pose Estimation

3.3.1 Problem Statement

We evaluate state-of-the-art 3D human pose estimation (HPE) models on our ASD-specific dataset to assess their robustness. HPE models, which process RGB images to generate skeletal representations, are vital in ASD research for analyzing repetitive or intense body movements. However, their performance is often limited by training on standard full-body captures, making them prone to errors in scenarios with occlusions or non-standard framing, as seen in ASD datasets. Studies have highlighted these limitations; for example, Muty and

Azizul [14] reported challenges in detecting arm flapping in unstructured data, and Vyas et al. [27] found reduced accuracy in recognizing behaviors over time in dynamic environments. Our goal is to benchmark HPE models on these challenging scenarios, identifying gaps in their applicability to ASD-focused research and providing direction for improved methodologies.

3.3.2 Methods Evaluated

We evaluated two state-of-the-art models, OpenPose [13] and MHFormer [10], on a GeForce RTX 3060 GPU. We use the original frames extracted from our videos that are JPEG images with a resolution of 1920×1080 pixels, ensuring that maximum accuracy is preserved. Downsampled images were avoided to maintain the fidelity of fine-grained movements essential for analyzing ASD-related behaviors. OpenPose uses Part Affinity Fields (PAFs) to detect and associate body keypoints, enabling accurate multi-person pose estimation even in complex scenarios. It processes high-resolution inputs effectively, allowing it to detect subtle repetitive movements such as head banging or hitting in ASD-specific data. MHFormer provides whole-body pose estimation with strong temporal consistency, accurately capturing detailed motion dynamics across frames. Its ability to handle occlusions and irregular movements makes it well-suited for ASD-focused studies.

3.3.3 Results

Since the pose estimation models were evaluated on videos without keypoint markers, the results are qualitative and based on visual inspection of the outputs as seen in Figure 3. Both OpenPose and MHFormer utilize pre-trained models that are optimized for standard, structured captures but perform surprisingly well on our dataset, which can be classified as “in-the-wild” for these models. OpenPose estimates 25 keypoints which includes fingers and eyes while MHFormer estimates 17 keypoints that represent the body. Both models demonstrate strong performance for activities involving head hitting with either hand, as these actions maintain visible keypoints for most of the frames. However, performance drops for head banging, where heavy occlusion prevent the fitting of most keypoints. Both models excel in detecting fully visible body parts but struggle with occluded areas, especially during intense or partial movements. OpenPose shows a notable advantage in multi-person scenarios, effectively separating keypoints for different individuals when more than one person is present in the frame. However, occlusions remain a challenge for both models. These limitations suggest vast opportunities for further ASD-specific research since the

capturing limitations in these cases call for models that are very flexible to occlusion.

4. Limitations and Future Work

As formerly mentioned, studies on the visual data of children with ASD are per se involved with challenges and concerns regarding data privacy, inclusiveness and diversity, and many more limitations regarding capturing process. While our study introduces the SIB-ASD dataset and establishes benchmarks for behavior classification and pose estimation, it is bound to notable limitations. In this section we are acknowledging a number of these limitations so to set a path for the future studies.

4.1. Data Size and Type

Computer vision research in the field of ASD pose and action and benefit from both image and video data. Our dataset focuses on images extracted from video captures. This method ensures that maximum number of frames are captured and they include a range of repeated action patterns. However, it compromises on the process of labeling. Hence, we managed to only extract 8,557 labeled images from the videos which may limit for modern deep learning methods that often rely on larger datasets. On a side note, an image dataset inherently limits the integration of temporal aspect of videos.

4.2. Diversity and Generalizability

The dataset consists of images of two participants, both of Hispanic ethnicity. This lack of demographic diversity raises concerns about potential biases and limits the generalizability of findings to broader populations. However, this limitation is inevitable, as the process of capturing image or video data requires the physical presence of subjects, inherently imposing demographic constraints. Building datasets that are more representative of diverse populations will require collaborative efforts to collect data from multiple sites around the world, which may take considerable time and resources. Studies like this serve as an important starting point to address the needs of specific patient groups and lay the groundwork for future research.

4.3. Quantitative analysis

Quantitative analysis in pose estimation relies on annotating keypoints, either using physical markers during capture or manual annotation tools. While markers may interfere with natural behavior, manual annotation is time-intensive, posing challenges for large-scale datasets. Future work can address these limitations by exploring automated annotation methods. Additionally, incorporating multimodal approaches, such as accelerometer signals or audio cues, can provide richer

context and improve recognition accuracy, particularly for occluded or partially visible actions.

4.4. Moral concerns and data privacy

Visual data captured from individuals, especially children with special conditions, is inherently sensitive and raises significant privacy concerns. As outlined in Section 3, informed consent was obtained from caregivers prior to data collection, ensuring ethical compliance. Additionally, a second round of consent was secured for presenting anonymized, face-blurred images to further protect participants' identities. However, ethical considerations extend beyond consent, encompassing issues such as data storage, access control, and the potential misuse of sensitive information.

4.5. Future Work

The field of computer vision for analyzing behaviors in children with ASD is still in its early stages, offering numerous opportunities for expansion in both data collection and model development. Based on the foundations laid by this study and the limitations encountered, we outline several directions for future research.

First, addressing the scarcity of data remains a critical goal. Future work can focus on capturing more annotated, full-body datasets that better represent ASD-related behaviors. Such datasets can aid supervised pose estimation and potentially shift focus toward pose-based datasets rather than image-based ones, improving privacy protection while preserving behavioral information. Expanding demographic diversity in these datasets will also enhance generalizability.

Second, integrating temporal aspects of behavior analysis can provide richer insights. While this study focuses on static images, future approaches can leverage sequential modeling techniques, such as recurrent neural networks (RNNs), temporal convolutional networks (TCNs), or transformers with spatio-temporal embeddings, to capture behavior progression over time.

Third, developing specialized model architectures tailored for ASD behavioral studies is another promising avenue. These models can be optimized for repetitive and intensive behaviors and benefit from multimodal data integration, such as accelerometer signals, to improve action detection and prediction.

Finally, addressing privacy and ethical concerns is key to enabling larger and more diverse datasets. Federated learning frameworks can be explored to train models on distributed datasets without exposing sensitive information. Additionally, pose-based datasets, rather than image datasets, can further mitigate privacy risks while enabling public data sharing for broader research.

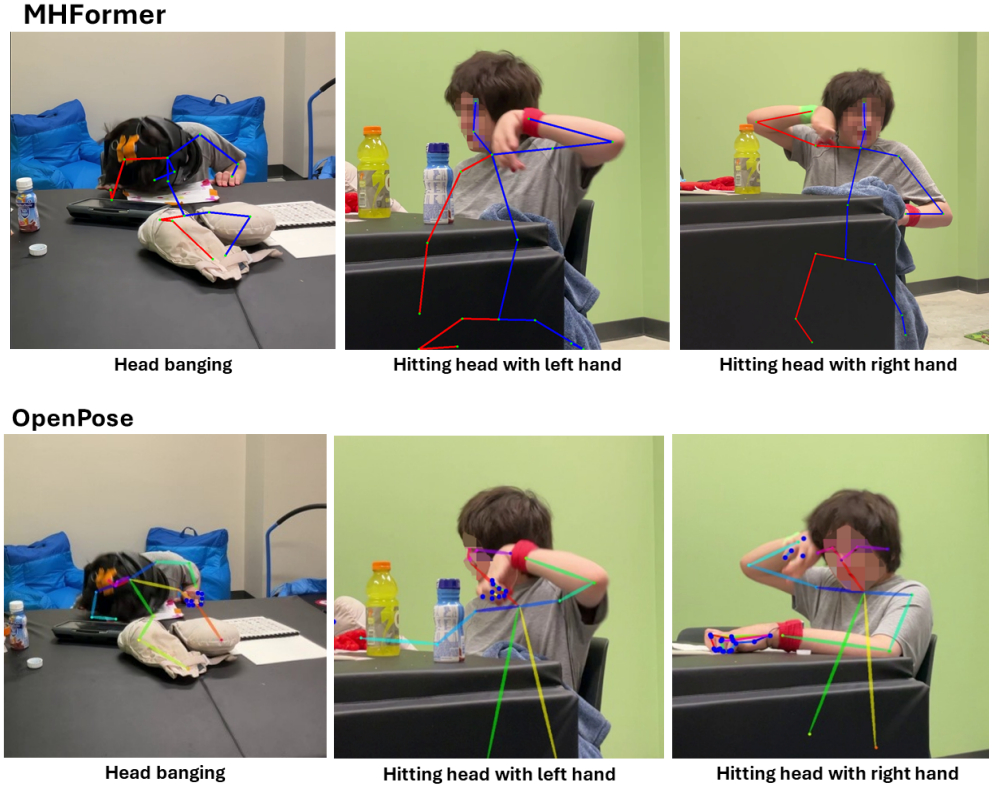


Figure 3. Results of applying MHFormer and Openpose on some representative frames of our data. The frames demonstrate the applicability of SOTA pose estimation models on scenarios with diverse poses such as head banging, hitting head with right hand, hitting head with left hand, and hitting head with both hands as well as different occlusion scenarios and camera angles.

5. Conclusion

Our work has introduced a specialized dataset called SIB-ASD for analyzing self-injurious behaviors (SIB) in individuals with autism spectrum disorder (ASD), providing a benchmark for advancing frame-level behavior classification and pose estimation in this domain. This dataset fills a critical gap by providing high-resolution, annotated frames capturing repetitive and intense body movements, such as head hitting and head banging. We evaluated state-of-the-art image classification and human pose estimation (HPE) methods, including OpenPose and AlphaPose, to establish benchmarks for behavior classification and pose recognition. Through training on 10 SOTA image classification models, we observed reasonable degrees of success in recognizing specific SIBs demonstrating the potentials of the dataset in representing repetitive nature of such behaviors. In the HPE evaluation, OpenPose and AlphaPose exhibited robust performance on high-resolution images, capturing body keypoints accurately under challenging conditions such as partial occlusions. However, limitations still emerge in scenarios with irregular movements, unstructured framing, and severe occlusion that are common in

ASD datasets. Our findings underscore the challenges of adapting general-purpose models to ASD datasets, emphasizing the need for specialized algorithms and full-body captures rather than sheer facial expression images. Moreover, multimodal approaches that integrate video data with complementary signals, such as accelerometer readings can benefit these studies. Future works should focus on the data gap for computer vision studies on children with ASD and developing specific action classification and pose estimation methods for more accurate diagnosis and behavioral analysis. This work contributes to the broader understanding of ASD-related behaviors and establishes a foundation for developing solutions that ensure better behavioral tracking and support for individuals with ASD.

6. Acknowledgements

This material is partially based upon work supported by the National Science Foundation under Grant No. 2153249. Any opinions, findings, and conclusions or recommendations expressed in this material are those of the author(s) and do not necessarily reflect the views of the National Science Foundation.

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